

Introduction

Two or more non-stationary time series are *cointegrated* if some linear combination of them is stationary. The classic example is interest rates at different maturities: each rate is non-stationary (random-walk-like), but spreads are mean-reverting. A vector error-correction model (VECM) captures the short-run dynamics of cointegrated series while respecting the long-run equilibrium relationship; the Johansen procedure tests how many cointegrating relations exist and estimates them jointly with the dynamic parameters.

Prerequisites

A working understanding of vector autoregressive (VAR) models, unit roots, and the I(1)/I(0) terminology for integrated and stationary series.

Theory

The VECM is

$$\Delta y_t = \Pi y_{t-1} + \sum_{i=1}^{p-1} \Gamma_i \Delta y_{t-i} + \varepsilon_t.$$

The matrix Π encodes the long-run relationships; its rank equals the number of cointegrating relations. Johansen's approach tests the rank via two statistics:

- **Trace test:** $\lambda_{\text{trace}}(r) = -T \sum_{i=r+1}^k \log(1 - \hat{\lambda}_i)$ tests $H_0 : \text{rank} \leq r$.
- **Maximum eigenvalue test:** $\lambda_{\text{max}}(r) = -T \log(1 - \hat{\lambda}_{r+1})$ tests $H_0 : \text{rank} = r$ vs $H_1 : \text{rank} = r + 1$.

The two often agree; when they disagree, the trace test is generally preferred. Once the rank is determined, the cointegrating vectors and adjustment speeds can be estimated.

Assumptions

Each series is individually I(1) (verifiable by ADF and KPSS); k -variate Gaussian innovations; the lag length p is correctly specified.

R Implementation

```
library(urca); library(vars)

data(Canada)
jo <- ca.jo(Canada, ecdet = "const", type = "eigen", K = 2, spec = "transitory")
summary(jo)

# Fit VECM with rank r = 1 (if supported by test)
vecm <- cajorls(jo, r = 1)
vecm
```

Output & Results

`ca.jo()` returns the eigenvalue and trace test statistics per candidate rank, along with critical values at 10/5/1 % levels. `cajorls()` fits the VECM at the chosen rank and returns the cointegrating vector and adjustment coefficients.

Interpretation

A reporting sentence: “Johansen’s trace test supported one cointegrating relationship in the Canada macroeconomic data (test statistic 32.8 vs 5 % critical value 25.3, $p < 0.05$); the corresponding cointegrating vector linked money demand and interest rates with adjustment speeds of -0.10 for money demand toward equilibrium.” Always report both trace and max-eigenvalue results, and the chosen lag length.

Practical Tips

- Confirm each series is $I(1)$ via ADF / KPSS before cointegration testing; cointegration is meaningful only between integrated series.
- The `ecdet` argument specifies whether deterministic terms (constant, trend) enter inside the cointegrating relation; choose based on visible long-run drift.
- Engle-Granger two-step is simpler for two series and gives essentially equivalent conclusions; Johansen generalises to k variables.
- Report both trace and max-eigenvalue tests; they typically agree but can disagree when one or two cointegrating relations are weak.
- For short samples (under 100 observations) and many variables, Johansen loses power; reduce the dimension or use bias-corrected critical values.
- Once the VECM is fitted, impulse-response functions and forecast-error variance decompositions on `vec2var(vecm)` extend the standard VAR diagnostic toolkit.

R Packages Used

`urca` for `ca.jo()`, `cajorls()`, and Johansen-related cointegration tests; `vars` for VAR modelling and `vec2var()` to convert a VECM back to VAR form for IRF analysis; `tsDyn::VECM()` for an alternative VECM implementation; `urca::ca.po()` for the Phillips-Ouliaris cointegration tests.

For Reviewers

What to look for in a paper using this method.

- Common misapplications.
- Diagnostics that should be reported but often aren’t.
- Red flags in tables and figures.
- What to verify.
- What an adequate Methods paragraph must contain.

See also — labs in this chapter

- Time series basics